Interpretability of black box predictive models

Seminar Presentation

Université du Québec à Montréal

August 13, 2025

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1 Context

2 Game theoretical approach

3 Model decomposition approach

4 Practical challenges

5 Conclusion

Ethical issues in worldwide insurance market

r/MtF • 3 yr. ago
halupadude

Has anyone seen a discount in auto insurance premium after changing your documents and gender marker to F?

[Discussion]

If so, was it automatic or did you have to ask the insurance company for it?

Ethical issues in worldwide insurance market

Ethical issues in worldwide insurance market

arXiv > cs > arXiv:2105.10174

Computer Science > Computers and Society

[Submitted on 21 May 2021]

Algorithmic Audit of Italian Car Insurance: Evidence of Unfairness in Access and Pricing

Alessandro Fabris, Alan Mishler, Stefano Gottardi, Mattia Carletti, Matteo Daicampi, Gian Antonio Susto, Gianmaria Silvello

We conduct an audit of pricing algorithms employed by companies in the Italian car insurance industry, primarily by gathering quotes through a popular comparison website. While acknowledging the complexity of the industry, we find evidence of several problematic practices. We show that birthplace and gender have a direct and sizeable impact on the prices quoted to drivers, despite national and international regulations against their use. Birthplace, in particular, is used quite frequently to the disadvantage of foreign-born drivers and drivers born in certain Italian cities. In extreme cases, a driver born in Laos may be charged 1,000 euros more than a driver born in Milan, all else being equal. For a subset of our sample, we collect quotes directly on a company website, where the direct influence of gender and birthplace is confirmed. Finally, we find that drivers with riskier profiles tend to see fewer quotes in the aggregator result pages, substantiating concerns of differential treatment raised in the past by Italian insurance regulators.

MORE FOR AUTO INSURANCE SOLELY BECAUSE OF THEIR GENDER

New legislation announced to improve equity!

Ethical issues in worldwide insurance market

Ethical issues in ~~worldwide insurance market~~ sensible fields

Black Box Model

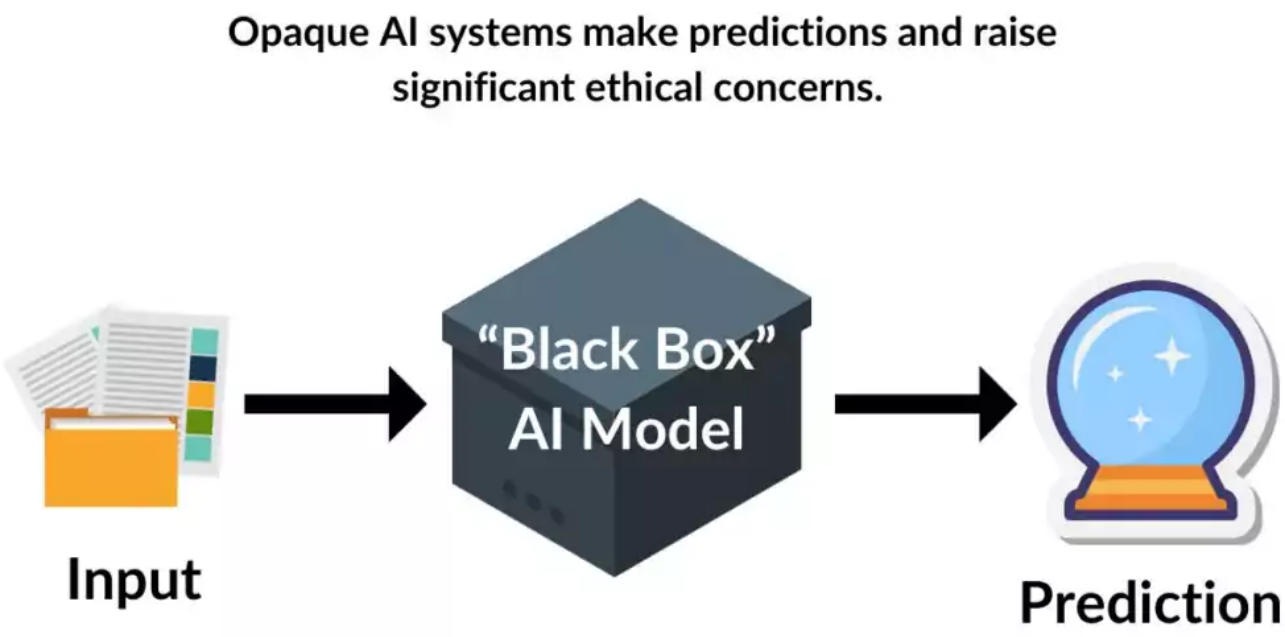

Figure 1
Black Box Model

Outline :

1. Context
2. Game theoretical approach
   - A story of poutines
   - Cooperative Game Theory and Machine Learning
   - Different types of allocations
3. Model decomposition approach
   - From different hypotheses...
   - ...to the same result ?
   - The results are actually different
4. Practical challenges
   - The project
   - Main challenges and our solutions
   - Computational Tips and Tricks
   - Results
5. Conclusion

1 Context

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- A story of poutines
- Cooperative Game Theory and Machine Learning
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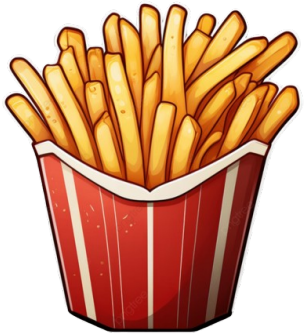
A story of poutines

- What if the cook forgets an ingredient ?

All is about poutine



3/10

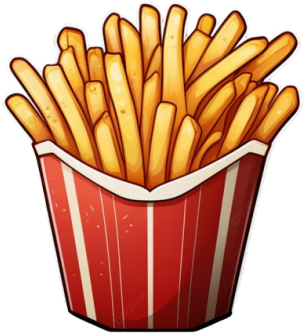
All is about poutine

1/10

All is about poutine

2/10

All is about poutine



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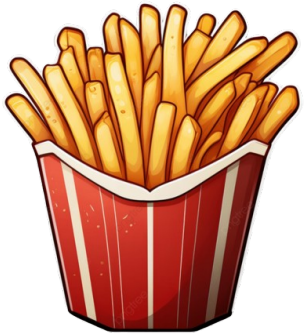
All is about poutine



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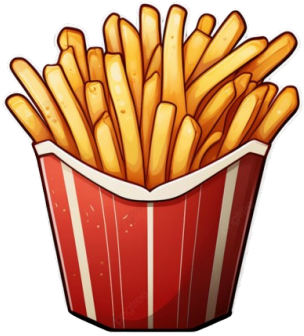
All is about poutine

4/10

All is about poutine



10/10

All is about poutine

0/10

How essential are the ingredients ?

			Grade
✗	✗	✗	0/10
✓	✗	✗	3/10
✗	✓	✗	1/10
✗	✗	✓	2/10
✓	✓	✗	7/10
✓	✗	✓	5/10
✗	✓	✓	4/10
✓	✓	✓	10/10

How essential are the variables ?

X_1	X_2	X_3	**Prediction Quality**
✗	✗	✗	0/10
✓	✗	✗	3/10
✗	✓	✗	1/10
✗	✗	✓	2/10
✓	✓	✗	7/10
✓	✗	✓	5/10
✗	✓	✓	4/10
✓	✓	✓	10/10

How essential are the variables ?

X_1	X_2	X_3	**Coalitional values**
✗	✗	✗	$v(\varnothing)$
✓	✗	✗	$v(\{1\})$
✗	✓	✗	$v(\{2\})$
✗	✗	✓	$v(\{3\})$
✓	✓	✗	$v(\{1,2\})$
✓	✗	✓	$v(\{1,3\})$
✗	✓	✓	$v(\{2,3\})$
✓	✓	✓	$v(\{1,2,3\})$

Why is it linked to Machine Learning ?

Let $f : X \to Y,\ Y \subseteq \mathbb{R}$ be a black-box model.

Let $X = (X_1, \ldots, X_d)$ be the **random inputs** of f.

Let $D = \{1, 2, \ldots, d\}$ be the index set of input features.

Let $\mathcal{P}_D$ be the powerset of D (the set of all the subsets of D).

Let $QoI : Y \to Q,\ Q \subseteq \mathbb{R}$ be the quantity of interest of the model :

$$QoI(f(X)) = f(X) \quad \text{or} \quad QoI(f(X)) = \mathbb{V}(f(X))$$

➢ We want to decompose $QoI(f(X))$ using the information stored in v.

How to choose v ?

Properties of the coalitional values

- $v : \mathcal{P}_D \to Q$ is the value measure
- $v(D) = QoI(f(X))$: The grand coalition build the total model

Examples of v

- $v(A) = \mathbb{E}[f(X)|X_A]$: Contribution of the features
- $v(A) = \mathbb{V}(\mathbb{E}[f(X)|X_A])$: Importance of the features

➢ **Arbitrary choice for game theorethical approach, we'll see after which one is the "right one"**

How to extract information from v ?

➢ We want to find a function ϕ which summarise the information stored in our values measures v and such that :

Two main decompositions :

- **Coalitional Decomposition :** $\phi : \mathcal{P}_D \to Q$

$$\Rightarrow QoI(f(X)) = \sum_{A \subseteq D} \phi(A)$$

- **Allocation (or Feature Decomposition) :** $\phi : D \to Q$

$$\Rightarrow QoI(f(X)) = \phi_0 + \sum_{i=1}^{d} \phi(i)$$

Back to the poutine example

Let say $QoI(f(X)) := f(X) = v(\{$, , $\}) = \mathbf{10/10}$: we want to decompose the $10/10$ grade.

➢ **Coalitional Decomposition:**

$10/10 = \phi(\varnothing)$

$+ \phi($ $) + \phi($ $) + \phi($ $)$

$+ \phi($, $) + \phi($, $) + \phi($, $)$

$+ \phi($, , $)$

➢ **Allocation :**

$10/10 = \phi_0 + \phi($ $) + \phi($ $) + \phi($ $)$

3 main steps

Step 1 : Choose a QoI to decompose

Example : $QoI(f(X)) = f(X) \implies v(A) = \mathbb{E}[f(X)|X_A]$

Step 2 : Compute the 2^d coalitional models

Example : $v(\{1,2\}) \leftarrow \mathbb{E}[f(X)|X_1, X_2])$

Step 3 : Aggregate information into d ϕ-values

Example : Shapley Values

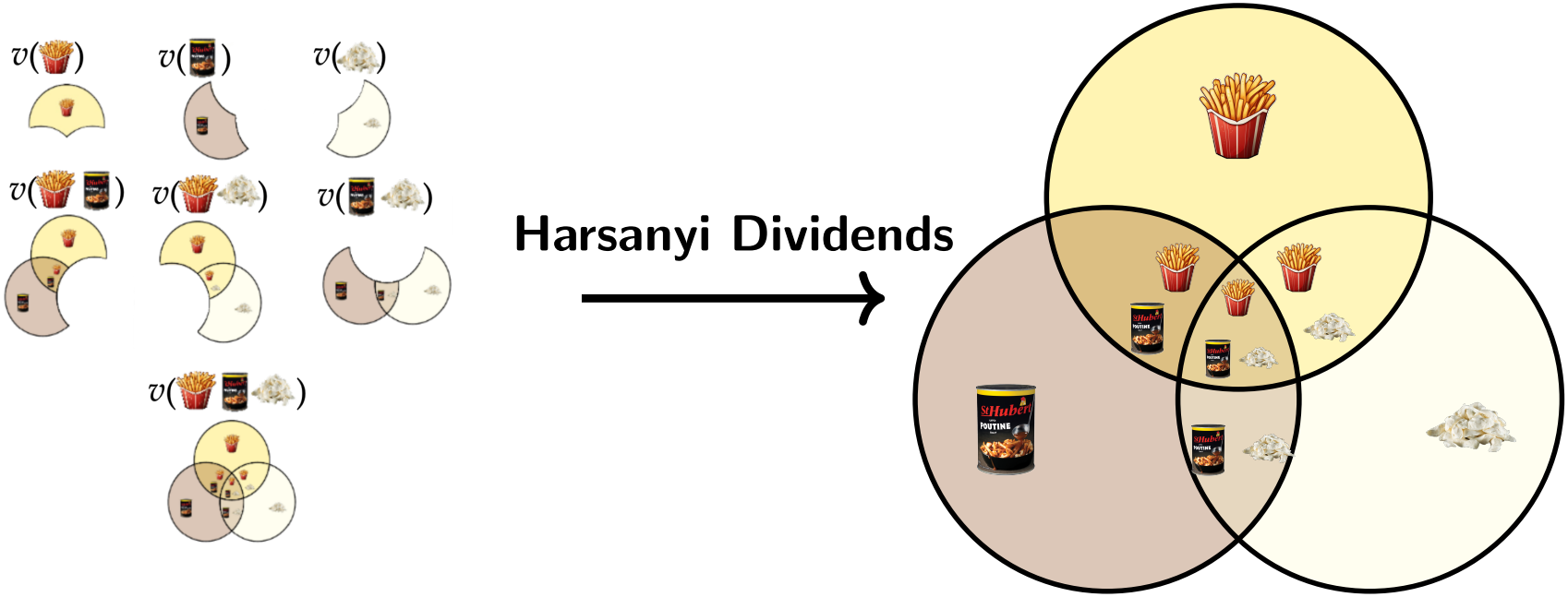
The Harsanyi Set

- 2 steps before the final allocation

$$\underbrace{v(A)}_{2^d} \xrightarrow{\text{Harsanyi Dividends}} \underbrace{\mathcal{D}(A)}_{2^d} \xrightarrow{\text{Harsanyi Allocation}} \underbrace{\phi^H(i)}_{d}$$

Step 1 : Harsanyi Dividends $v(A) \rightarrow \mathcal{D}(A)$

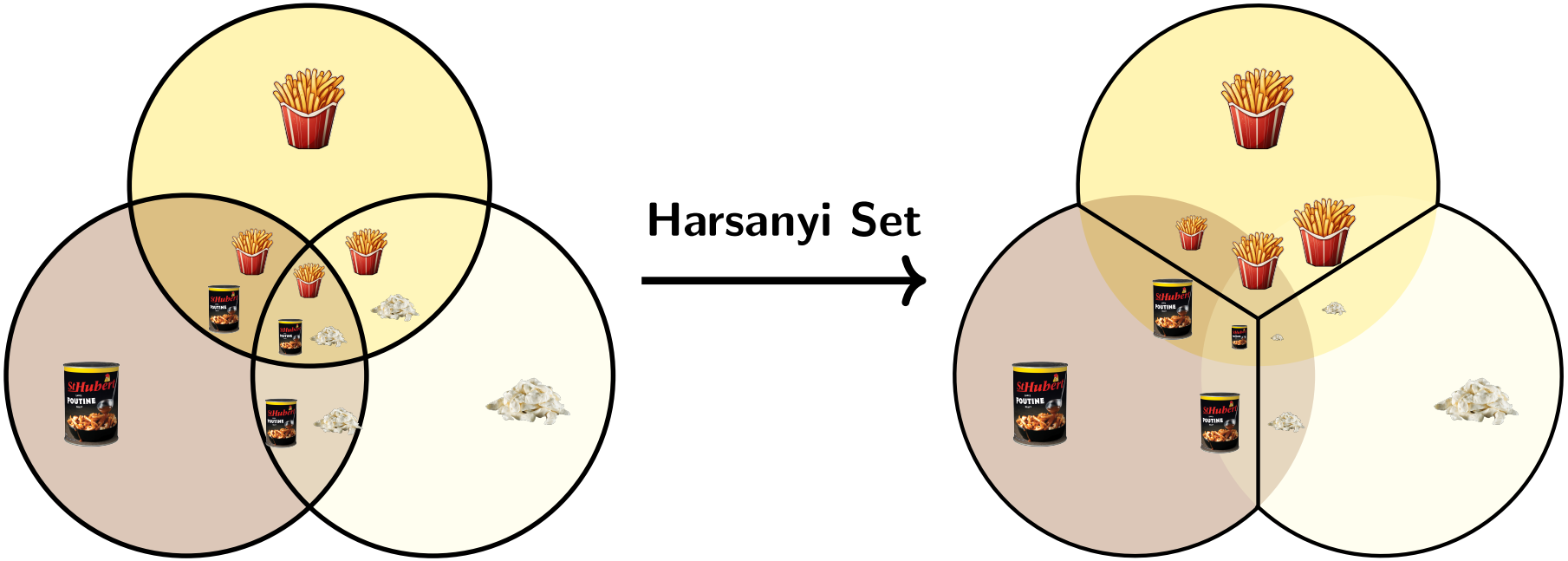
➢ "The Inclusion-Exclusion principle"



$$\mathcal{D}(A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} v(B), \ \forall A \in \mathcal{P}_D$$

Step 2 : Harsanyi Allocation $\mathcal{D}(A) \rightarrow \phi^H(i)$

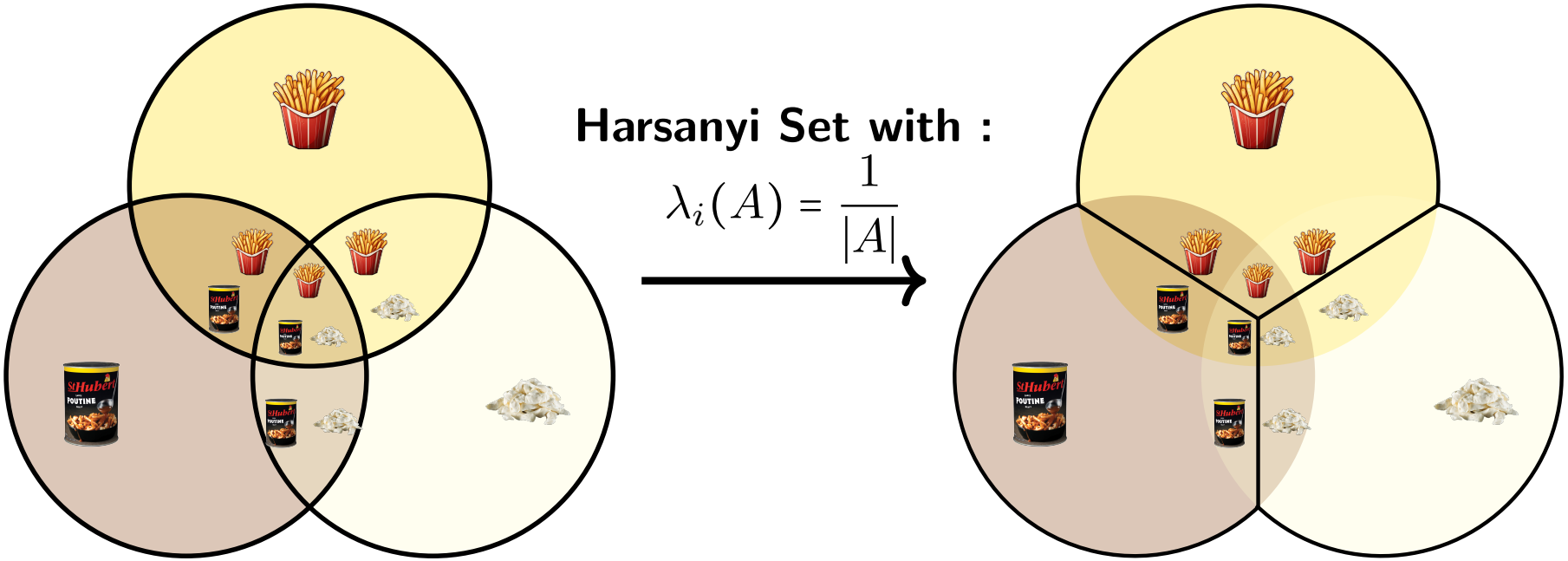
- "The weighted sum of the Harsanyi Dividends"



$$\phi^H(i) = \sum_{A \in \mathcal{P}_D \,:\, i \in A} \lambda_i(A)\mathcal{D}(A), \quad \forall i \in D$$

The Shapley Values : A particular case of the Harsanyi Set

- The Shapley Values : egalitarian repartition of the added-values



$$\phi^S(i) = \sum_{A \in \mathcal{P}_D \,:\, i \in A} \frac{1}{|A|} \mathcal{D}(A), \ \forall i \in D$$

Back to the poutine example : Harsanyi Dividends

			$v(A)$	$\mathcal{D}(A)$
✗	✗	✗	0/10	0/10
✓	✗	✗	3/10 =	3/10 =
✗	✓	✗	1/10 =	1/10 =
✗	✗	✓	2/10 =	2/10 =
✓	✓	✗	7/10 =	3/10 =
✓	✗	✓	5/10 =	0/10 =
✗	✓	✓	4/10 =	1/10 =
✓	✓	✓	10/10 =	0/10 =

Back to the poutine example : Harsanyi Egalitarian Set

			$v(A)$	$\frac{1}{\lvert A\rvert}\mathcal{D}(A)$
✗	✗	✗	$0/10$	$0/10$
✓	✗	✗	$3/10 =$	$3/10 =$
✗	✓	✗	$1/10 =$	$1/10 =$
✗	✗	✓	$2/10 =$	$2/10 =$
✓	✓	✗	$7/10 =$	$1.5/10 =$ $/2$
✓	✗	✓	$5/10 =$	$0/10 =$ $/2$
✗	✓	✓	$4/10 =$	$0.5/10 =$ $/2$
✓	✓	✓	$10/10 =$	$0/10 =$ $/3$

Back to the poutine example : Harsanyi Egalitarian Set

i	$\phi(i)$
	$3/10 =$ $+$ $/2 +$ $/2 +$ $/3$
	$2.5/10 =$ $+$ $/2 +$ $/2 +$ $/3$
	$4.5/10 =$ $+$ $/2 +$ $/2 +$ $/3$

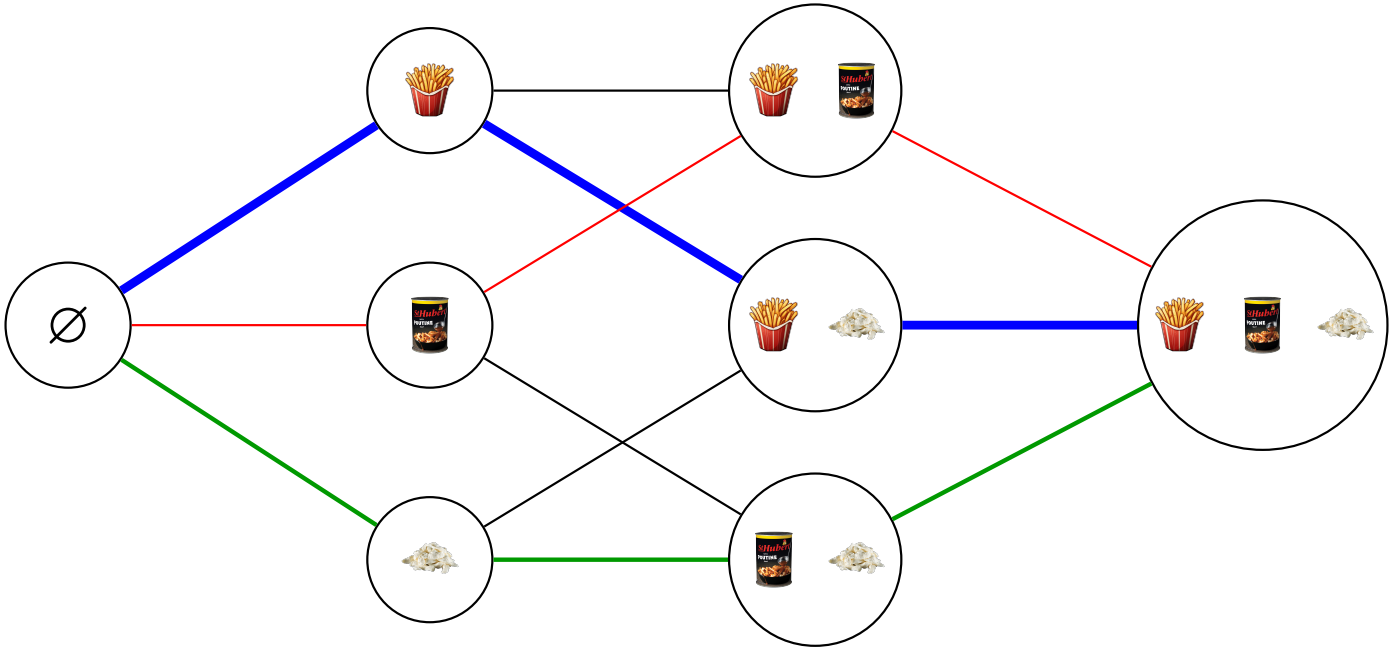
The Weber Set

- Direct allocation based on order importance

$$\underbrace{v(A)}_{2^d} \xrightarrow{\text{Weber Allocation}} \underbrace{\phi^W(i)}_{d}$$

Back to the Poutine Example : Weber Set

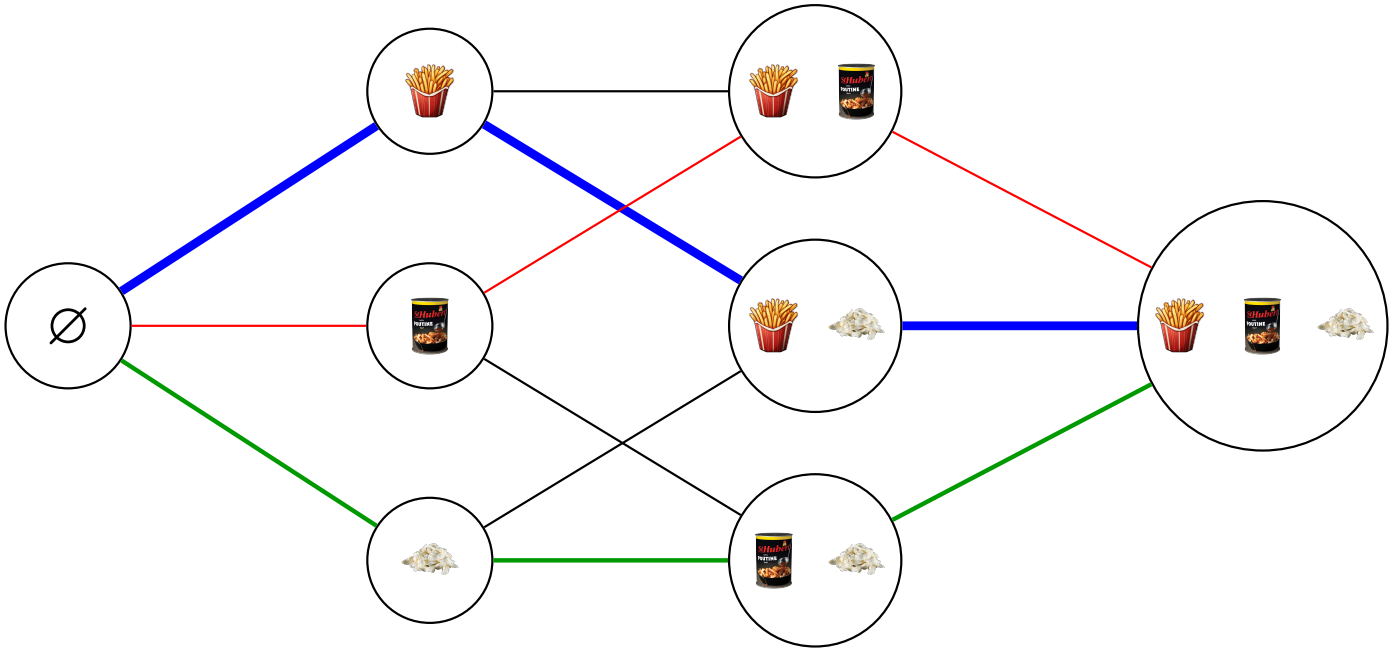
- "Order matters"



Some ingredients are more likely used in first

Back to the Poutine Example : Weber Set

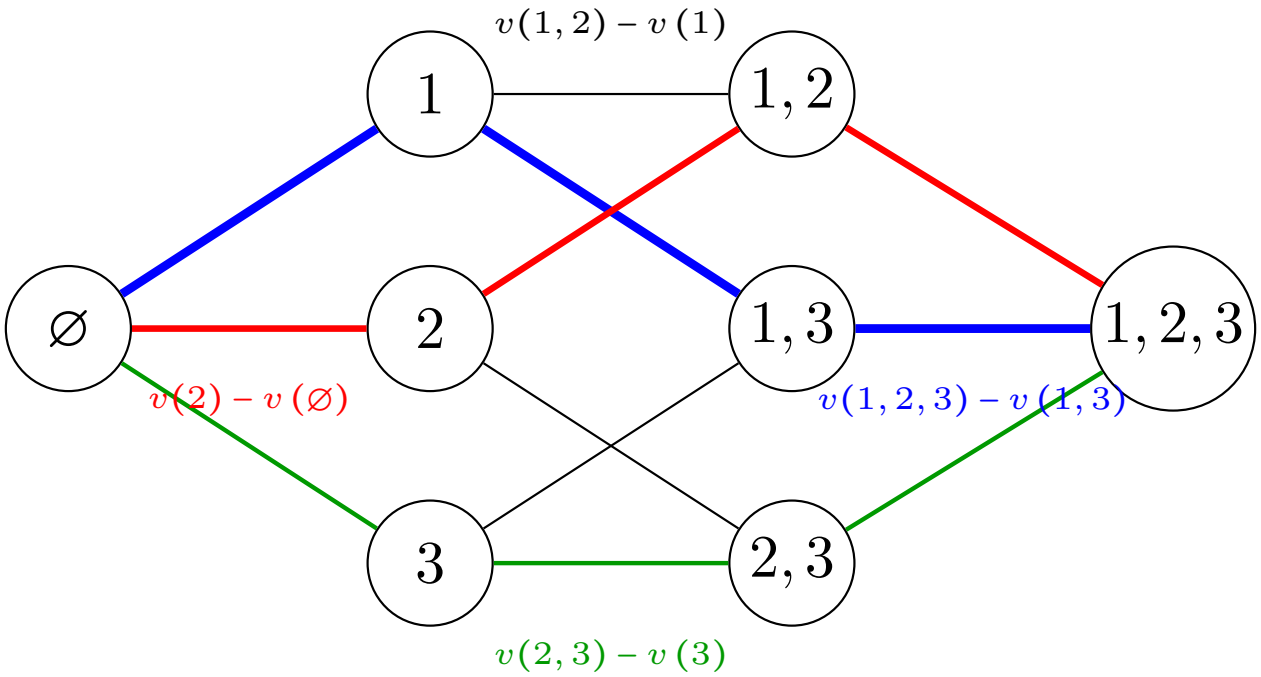
- "Order matters"



$v($ $) - v(\varnothing)$ << $v($, , $) - v($, $)$

The Weber Set

- "The marginal contribution expectation based on order"



$$\boxed{\phi^W(i) = \mathbb{E}_{\pi \sim p}\left[v\left(C_{\pi(i)}(\pi)\right) - v\left(C_{\pi(i)-1}(\pi)\right)\right], \ \forall i \in D}$$

The Shapley Values : A particular case of the Weber Set

Weber Set with uniform discrete distribution

$$\phi^S(i) = \sum_{\pi \in \Pi_D} \frac{1}{d!} \left[v\left(C_{\pi(i)}(\pi)\right) - v\left(C_{\pi(i)-1}(\pi)\right) \right], \ \forall i \in D$$

- This allocation correspond to the Shapley Values

Marginal Contribution Formulas

- **We can express all the 3 allocations as a weighted sum of marginal contributions :**

$$\boxed{\phi(i) = \sum_{A \in \mathcal{P}_D \,:\, i \in A} \beta(i, A) \left[v(A) - v(A \setminus i) \right], \ \forall i \in D}$$

- Shapley : $\beta(i, A) = \dfrac{(|A| - 1)!(|D| - |A|)!}{|D|!}$
- Harsanyi : $\beta(i, A) = \sum_{B \supseteq A} (-1)^{|A| - |B|} \lambda_i(B)$
- Weber : $\beta(i, A) = \mathbb{P}_p\{\pi^i = A\}$

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3 Model decomposition approach
- From different hypotheses...
- ...to the same result ?
- The results are actually different

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The Hoeffding decomposition

- In the previous part, we chose the value measure v arbitrarily (ex. $v(A) = \mathbb{E}[f(X)|X_A]$)
- We don't know if it will lead to a "good" decomposition
- We now want to find the mathematically "right" value measure v
- We rely on **Hoeffding's Decomposition Theorem** from functional analysis

How to (really) choose v

Hoeffding's decomposition Theorem

Let $f(X) \in \mathbb{L}^2(\sigma(X))$.
There exists a unique decomposition of the form :

$$f(X) = \sum_{A \in \mathcal{P}_D} f_A(X_A) \quad \text{a.s.}$$

The Hoeffding decomposition

The Möbius Transform for independent features

Under the **mutually independent** assumption :

$$f_A(X_A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} \mathbb{E}[f(X)|X_B], \quad \text{i.e :}$$

$$f(X) = \sum_{A \in \mathcal{P}_D} \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} \mathbb{E}[f(X)|X_B] \quad \text{a.s.}$$

- Harsanyi Dividends $\mathcal{D}$ Coalitional Decomposition with the choice $v(B) = \mathbb{E}[f(X)|X_B]$

What if the features are not independant

The Möbius Transform for non independent features

Let $f(X) \in \mathbb{L}^2(\sigma(X))$.
There exists a unique decomposition of the form :

$$f(X) = \sum_{A \in \mathcal{P}_D} f_A(X_A) \quad \text{a.s.}$$

with :

$$f_A(X_A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} \mathbb{M}_B$$

with $\mathbb{M}_B$ the oblique projection of $f(X)$ onto $\mathbb{L}^2(\sigma_B)$

All is about assumptions

- **Game Theorethical Approach :**
  No assumptions, decomposition works for every choice of v
  $\implies$ Poor mathematical argument, arbritary choice of v

- **Model Decomposition Approach :**
  Strong Assumptions, we control the decomposition
  $\implies$ Strong mathematical argument, give a framework for the choice of v

However, in practice, the choice of v is the conditionnal expectation $\implies$ unaccurate decompositions

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4 Practical challenges
- The project
- Main challenges and our solutions
- Computational Tips and Tricks
- Results

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Goals of the project

Mains Goals

- Write a R package proposing different methods of black-box models interpretability
- The methods should be "fast", optimized and compatible with R ML objects

Challenges

- **Time complexity** of the different methods : $o(nd2^d)$ or more depending on the method
- **Space complexity** : manipulation of 2^d models : too heavy in memory
- `R` **slowingness**

R slowingness : a solution

- Interfacing R and C++ with Rcpp

R ⟺ Rcpp C++

Time complexity : a solution

- **Shapley, Harsanyi, Weber** : adaptative computing parallelization
- **Weber** : Sampling estimation for highest values of d (complexity of $o(n!)$)

Space complexity : a solution

Step 1 : Training

We train the 2^d coalitional models and store them on hard disk

▼

Step 2 : Computing of v

We compute the 2^d coalitional values and store them on hard disk

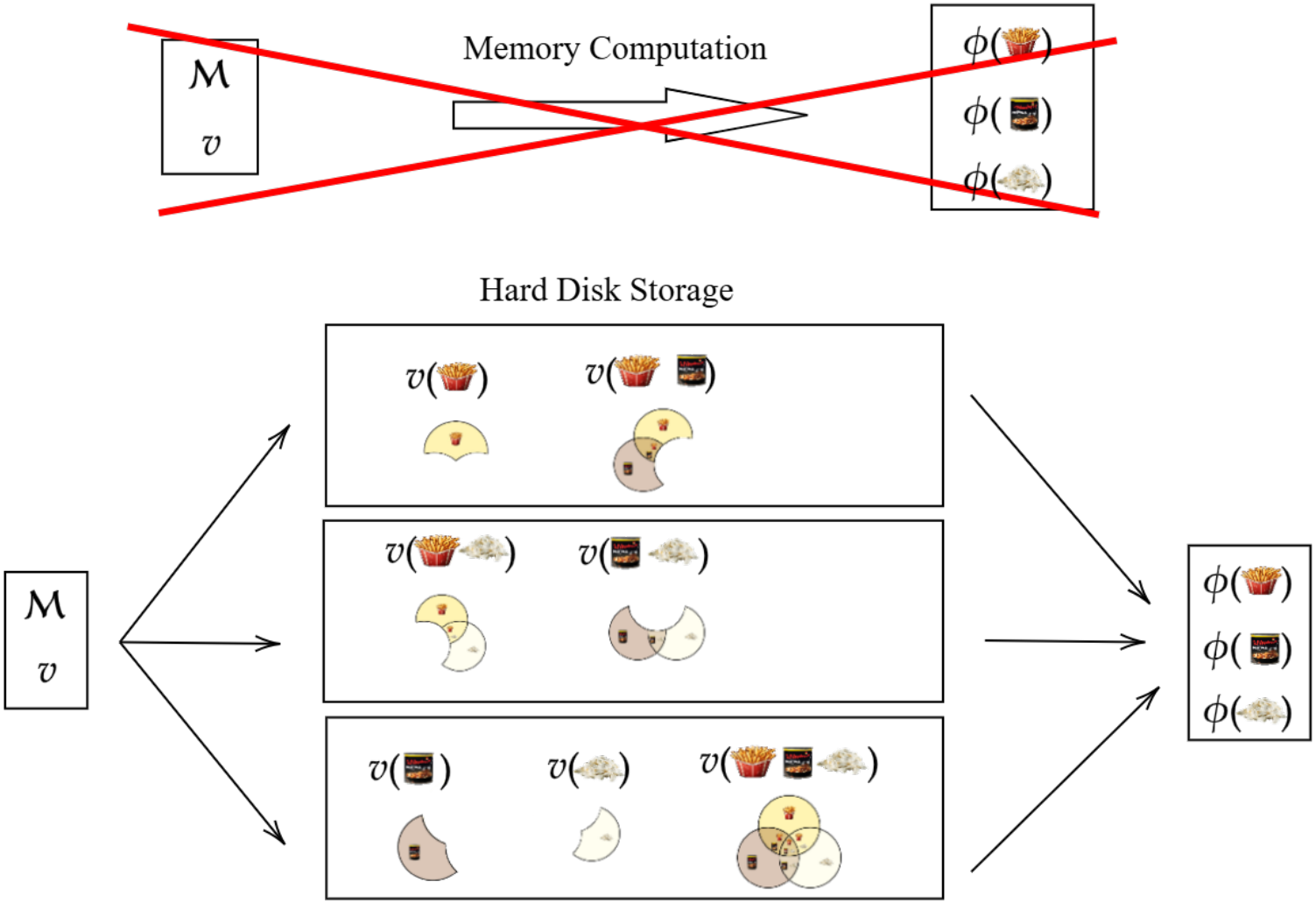
▼

Step 3 : Allocation

We compute the allocations by accessing the coalitional values on hard disk storage

➢ **Make different allocations without re-training the models every time**

Space and time complexity : a solution



How to tidy up the models on the hard disk ?

- We need to index the models an optimal way so we can access any quickly

Lexicographical Ordering

Maps each coalition to a unique integer by converting the base 2 bit-representation of the coalitions into a base 10 number.

Example : Coalition $\{1, 2, 4\}$ (out of $d = 5$ features)

$$[1 \quad 1 \quad 0 \quad 1 \quad 0] \quad \Rightarrow \quad 1 \cdot 2^0 + 1 \cdot 2^1 + 0 \cdot 2^2 + 1 \cdot 2^3 + 0 \cdot 2^4 = \mathbf{11}$$

- **In `C++`, checking the active bits of a binary number is native**

How to tidy up the models on the hard disk ?

- We can't store all the models on one file (could lead to parallelization issues and too heavy files)
- We need to "separate" models on different files

Separation of Models in Blocks

2 different approaches :

- Storage per sequential blocs
- Storage per coalitional properties

Storage per sequential blocs

- Sequential bloc storage by lexicographical order

1	2	3	4	5
1	4	7	10	13
2	5	8	11	14
3	6	9	12	15

$\boxed{0}$

Example for $d = 4$

Storage per sequential blocs

- Sequential bloc storage by lexicographical order

1	2	3	4	5
$\{1,0,0,0\}$	$\{0,0,1,0\}$	$\{1,1,1,0\}$	$\{0,1,0,1\}$	$\{1,0,1,1\}$
$\{0,1,0,0\}$	$\{1,0,1,0\}$	$\{0,0,0,1\}$	$\{1,1,0,1\}$	$\{0,1,1,1\}$
$\{1,1,0,0\}$	$\{0,1,1,0\}$	$\{1,0,0,1\}$	$\{0,0,1,1\}$	$\{1,1,1,1\}$

Example for $d = 4$

Storage per sequential blocs

- Sequential bloc storage by lexicographical order

1	2	3	4	5
$v\{1\}$	$v\{3\}$	$v\{1,2,3\}$	$v\{2,4\}$	$v\{1,3,4\}$
$v\{2\}$	$v\{1,3\}$	$v\{4\}$	$v\{1,2,4\}$	$v\{2,3,4\}$
$v\{1,2\}$	$v\{2,3\}$	$v\{1,4\}$	$v\{3,4\}$	$v\{1,2,3,4\}$

$\boxed{v\{\varnothing\}}$

Example for $d = 4$

Storage per sequential blocs

Advantages

- Flexible size of blocs and memory managment
- Parallelization is easy to implement (blocs are already the same size)
- Good indexation properties (We directly know the coalitional binary number)

Storage per coalitional properties

- Coalitions are stored depending on their first player and their number of players. Inside the cells, they are sorted by lexicographical order

	1	2	3	4
1	1	3 5 9	7 11 13	15
2	2	6 10	14	∅
3	4	12	∅	∅
4	8	∅	∅	∅

$\boxed{0}$

Example for $d = 4$

Storage per coalitional properties

- Coalitions are stored depending on their first player and their number of players. Inside the cells, they are sorted by lexicographical order

	1	2	3	4
1	$\{1,0,0,0\}$	$\{1,1,0,0\}$ $\{1,0,1,0\}$ $\{1,0,0,1\}$	$\{1,1,1,0\}$ $\{1,1,0,1\}$ $\{1,0,1,1\}$	$\{1,1,1,1\}$
2	$\{0,1,0,0\}$	$\{0,1,1,0\}$ $\{0,1,0,1\}$	$\{0,1,1,1\}$	$\varnothing$
3	$\{0,0,1,0\}$	$\{0,0,1,1\}$	$\varnothing$	$\varnothing$
4	$\{0,0,0,1\}$	$\varnothing$	$\varnothing$	$\varnothing$

$\boxed{\{0,0,0,0\}}$

Example for $d = 4$

Storage per coalitional properties

- Coalitions are stored depending on their first player and their number of players. Inside the cells, they are sorted by lexicographical order

	1	2	3	4
1	$v\{1\}$	$v\{1,2\}$ $v\{1,3\}$ $v\{1,4\}$	$v\{1,2,3\}$ $v\{1,2,4\}$ $v\{1,3,4\}$	$v\{1,2,3,4\}$
2	$v\{2\}$	$v\{2,3\}$ $v\{2,4\}$	$v\{2,3,4\}$	$\varnothing$
3	$v\{3\}$	$v\{3,4\}$	$\varnothing$	$\varnothing$
4	$v\{4\}$	$\varnothing$	$\varnothing$	$\varnothing$

$\boxed{v\{\varnothing\}}$

Example for $d = 4$

Storage per coalitional properties

Advantages

- We have useful informations about the coalitions in each cell (we know that the first players doesn't play)
- Parallelization is still possible if we distribute smartly the work

Storage per coalitional properties

Drawbacks

- A fixed size of blocs : $\frac{d(d+1)}{2}$ blocs could involve memory problems and unbalance
- Find the coalitional binary numbers in each cell could be costly (compared to storage per sequential blocs)

Splited marginal contribution

- **We have shown that we can write the allocations as a weighted sum of marginal contributions :**

$$\boxed{\phi(i) = \sum_{A \in \mathcal{P}_D \,:\, i \in A} \beta(i, A) \left[v(A) - v(A \setminus i) \right], \ \forall i \in D}$$

SMC Algorithm

- **For** A **in** $\mathcal{P}_D$
  - **If** $i \in A$: $\phi(i) \leftarrow \phi(i) + \beta(i, A) v(A)$
  - **If** $i \notin A$: $\phi(i) \leftarrow \phi(i) - \beta(i, A \cup i) v(A)$

A key algorithm : Fast Möbius Transform

- **Sometimes, we need to explicitly compute the Harsanyi Dividends :**

$$\boxed{\mathcal{D}(A) = \sum_{B \in \mathcal{P}_A} (-1)^{|A|-|B|} v(B), \ \forall A \in \mathcal{P}_D}$$

FMT Algorithm : a recursive algorithm

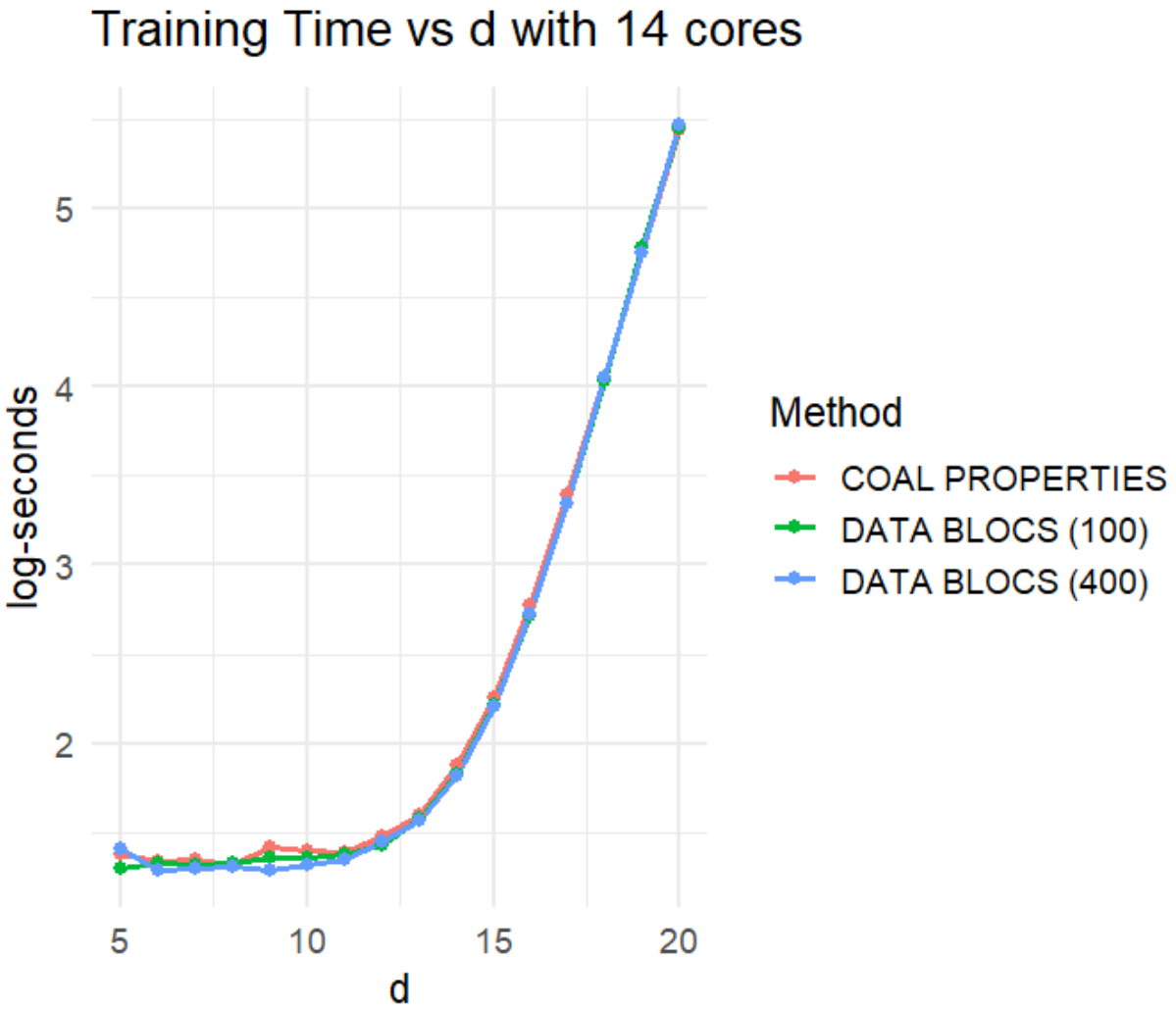
- **For** i **in** D
  - **For** A **in** $\mathcal{P}_D$
    - **If** $i \in A$: $v(A) \leftarrow v(A) - v(A \setminus \{i\})$

Setup

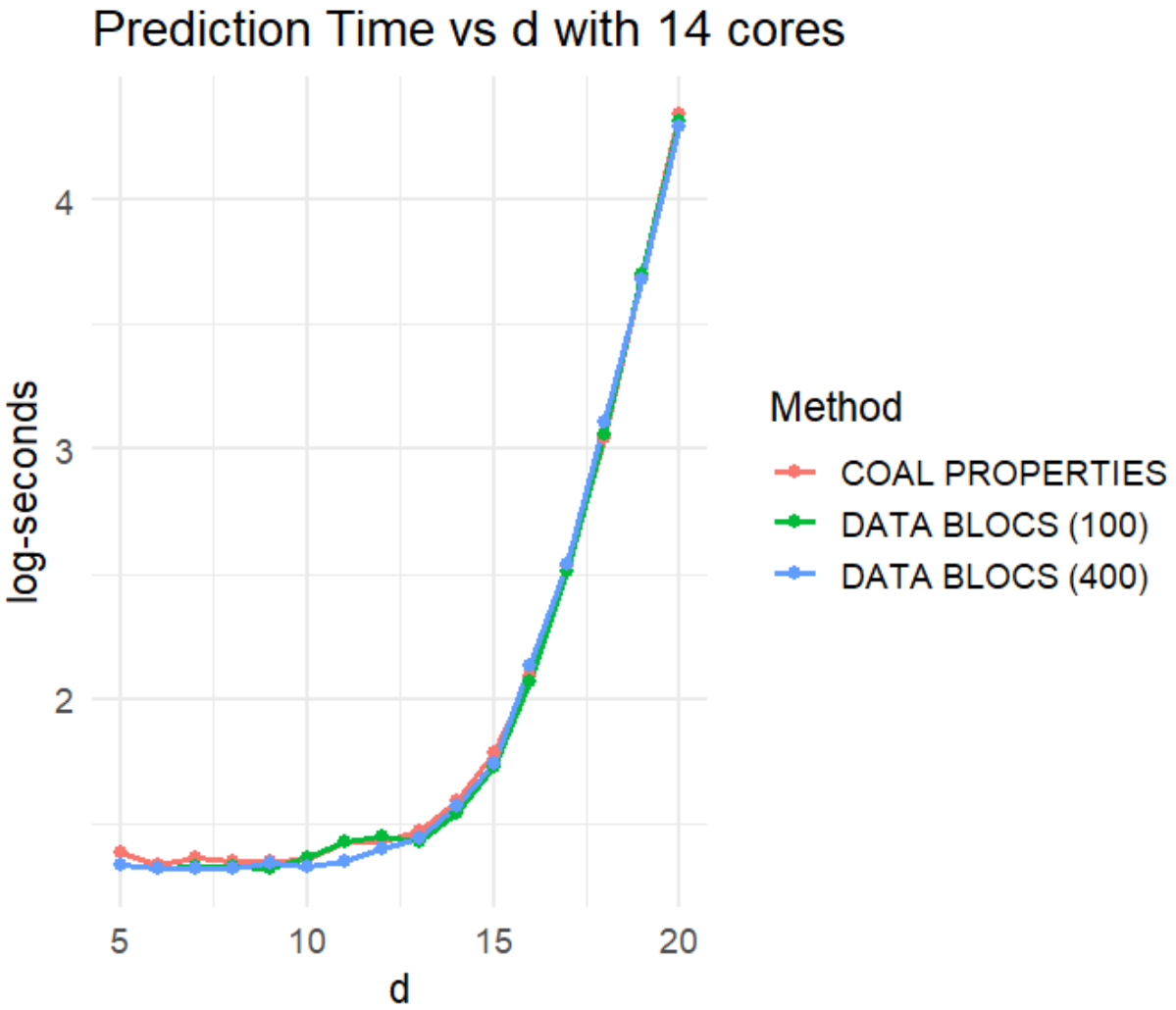
➢ **We tried the solutions and measured the time cost for :**

- Linear model
- $d = 5$ to 20 variables (gaussian)
- $n = 5000$ individuals
- Training, Prediction, Shapley, Harsanyi
- Sequential Blocs storage (max 100 and max 400 blocs), Coalitionnal Properties storage
- 14 virtual hearts, 32 GB memory

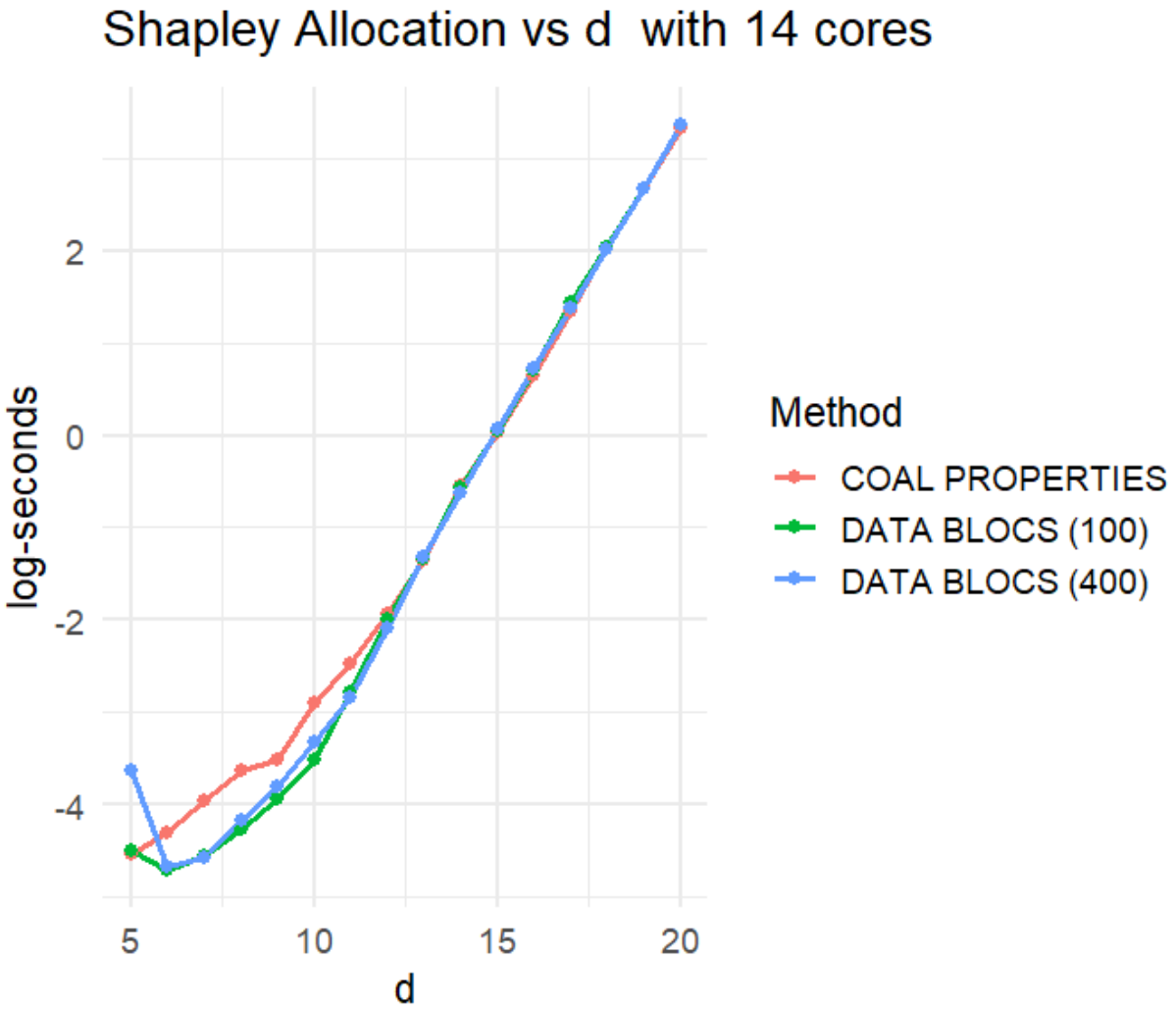
Training Time



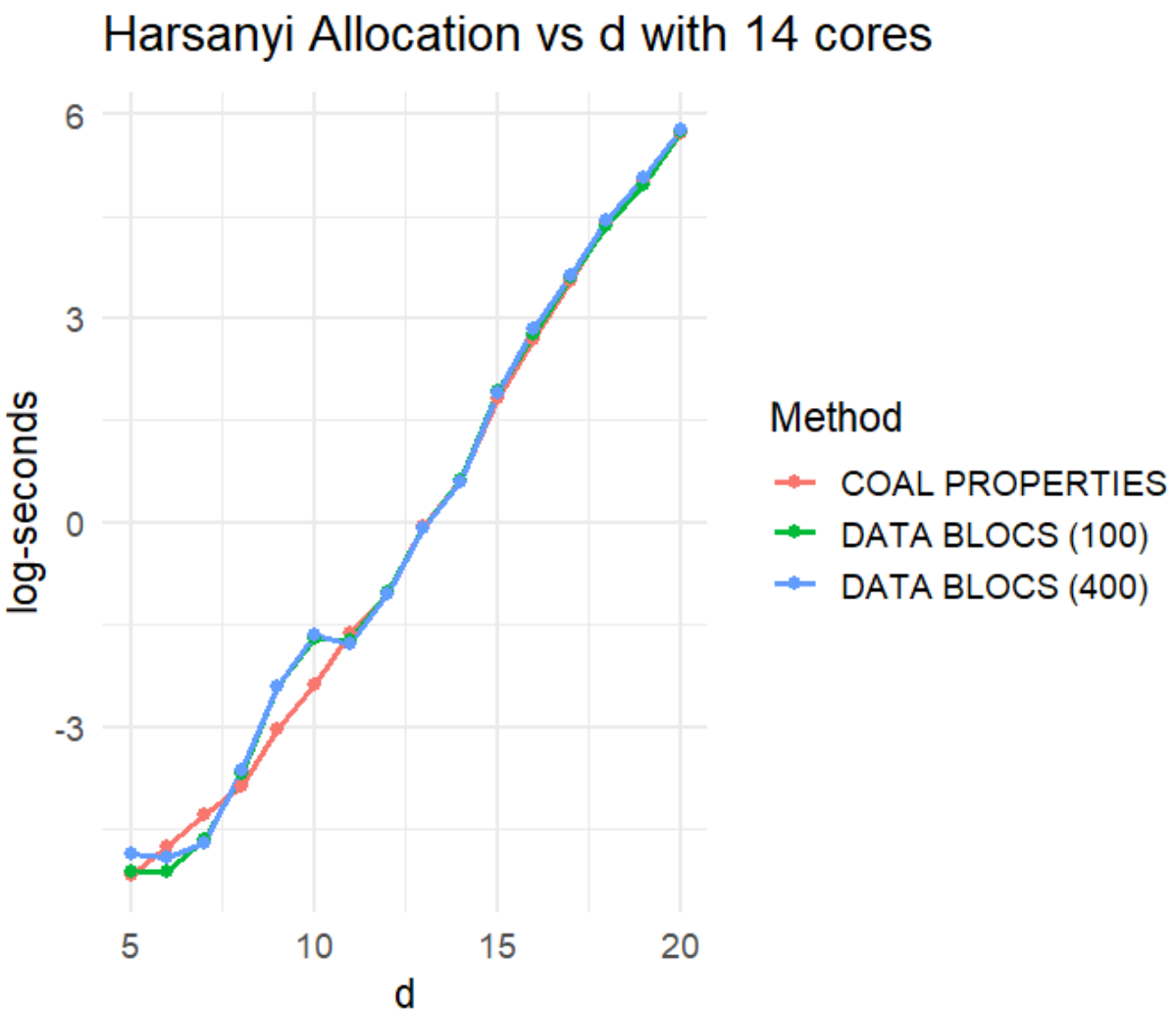
Prediction Time



Shapley Allocation



Harsanyi Allocation



1 Context

2 Game theoretical approach

3 Model decomposition approach

4 Practical challenges

5 Conclusion

What did you learn today ?

- **Game theoretical approach** $\neq$ **Model Decomposition approach**
- **Shapley, Harsanyi, Weber** allocations are linked together by the marginal contribution formula
- Computations are costly $\Longrightarrow$ We need to distribute the tasks and use the hard disk

Following of the project and areas for improvement

- **Computationally :**
  Find a way to store the minimum requirement for each model (i.e. randomforest, xgboost)

- **Theoretically :**
  Find closed formulas for Harsanyi and Weber marginal contribution coefficients $\beta(i, A)$ for specific choices of weight systems λ and permutation distributions p

Thanks

THANK YOU FOR LISTENING